Appendix A. Detailed Numerical Results of Nowcasting Effectiveness

Table A.1: Predictive performances on the test set evaluated using classification metrics (CSI and F1 scores, where higher values indicate better performance). MultiAir performs best across all pollution levels and lead times. In each setting, the best and the second-best performances are in **bold** and underlined.

Pollution level	Lead time	MultiAir CSI	MultiAir F1	ASOC+ CSI	ASOC+ F1	LSTM CSI	LSTM F1	Transformer CSI	Transformer F1	TimesNet CSI	TimesNet F1	TimeMixer CSI	TimeMixer F1	AirFormer CSI	AirFormer F1
Severity 2 or worse ($> 15\mu g/m^3$)	1 hr.	**0.822**	**0.902**	0.720	0.837	0.804	0.891	0.793	0.885	0.697	0.822	0.765	0.867	0.755	0.861
	2 hr.	**0.786**	**0.880**	0.707	0.828	0.755	0.861	0.752	0.859	0.687	0.815	0.735	0.848	0.694	0.819
	3 hr.	**0.757**	**0.862**	0.689	0.816	0.718	0.836	0.718	0.836	0.677	0.807	0.711	0.831	0.646	0.785
	4 hr.	**0.732**	**0.845**	0.670	0.802	0.690	0.816	0.688	0.815	0.664	0.798	0.688	0.815	0.596	0.747
	5 hr.	**0.708**	**0.829**	0.651	0.788	0.664	0.798	0.663	0.797	0.652	0.798	0.667	0.800	0.557	0.716
	6 hr.	**0.688**	**0.815**	0.631	0.774	0.644	0.784	0.641	0.782	0.640	0.781	0.651	0.788	0.521	0.685
Severity 3 or worse ($> 35\mu g/m^3$)	1 hr.	**0.756**	**0.861**	0.567	0.724	0.726	0.842	0.680	0.810	0.562	0.720	0.646	0.785	0.654	0.791
	2 hr.	**0.686**	**0.814**	0.542	0.703	0.628	0.772	0.614	0.761	0.543	0.704	0.594	0.745	0.569	0.725
	3 hr.	**0.633**	**0.775**	0.514	0.679	0.554	0.713	0.558	0.717	0.519	0.683	0.551	0.711	0.498	0.665
	4 hr.	**0.589**	**0.741**	0.487	0.655	0.498	0.665	0.511	0.677	0.490	0.658	0.513	0.678	0.443	0.614
	5 hr.	**0.552**	**0.711**	0.459	0.629	0.454	0.624	0.472	0.641	0.466	0.636	0.483	0.651	0.392	0.563
	6 hr.	**0.519**	**0.683**	0.431	0.603	0.419	0.590	0.438	0.610	0.446	0.617	0.459	0.629	0.338	0.505
Severity 4 ($> 75\mu g/m^3$)	1 hr.	**0.699**	**0.823**	0.322	0.487	0.648	0.787	0.561	0.719	0.305	0.467	0.537	0.699	0.638	0.779
	2 hr.	**0.601**	**0.751**	0.287	0.446	0.505	0.671	0.487	0.655	0.273	0.429	0.465	0.634	0.501	0.668
	3 hr.	**0.527**	**0.691**	0.251	0.401	0.405	0.576	0.424	0.595	0.226	0.369	0.405	0.576	0.397	0.568
	4 hr.	**0.471**	**0.641**	0.217	0.357	0.339	0.506	0.372	0.542	0.187	0.316	0.344	0.512	0.314	0.478
	5 hr.	**0.418**	**0.590**	0.186	0.314	0.287	0.446	0.327	0.493	0.158	0.273	0.306	0.468	0.240	0.388
	6 hr.	**0.369**	**0.539**	0.157	0.217	0.232	0.377	0.289	0.449	0.144	0.251	0.276	0.433	0.176	0.300
Average		**0.629**	**0.764**	0.472	0.614	0.554	0.698	0.555	0.702	0.463	0.608	0.544	0.693	0.496	0.648

Tables A.1 and A.2 report the detailed numerical values used for Fig-

Table A.2: Predictive performances on the test set evaluated using regression metrics (RMSE and MAE, where lower values indicate better performance). MULTIAIR performs best across all lead times. In each setting, the best and the second-best performances are in **bold** and underlined.

Lead	**MULTIAIR**		ASOC+		LSTM		Transformer		TimesNet		TimeMixer		AirFormer	
time	RMSE	MAE	RMSE	MAE	RMSE	MAE	RMSE	MAE	RMSE	MAE	RMSE	MAE	RMSE	MAE
1 hr.	**3.998**	**2.727**	7.180	4.734	4.539	3.051	5.182	3.427	7.431	5.139	5.821	3.871	5.692	4.277
2 hr.	**5.184**	**3.440**	7.601	4.990	6.146	4.010	6.407	4.161	7.817	5.389	6.842	4.509	7.013	5.026
3 hr.	**6.108**	**3.992**	8.111	5.298	7.308	4.727	7.455	4.801	8.249	5.644	7.697	5.029	8.132	5.694
4 hr.	**6.886**	**4.463**	8.624	5.609	8.184	5.280	8.366	5.373	8.750	5.944	8.483	5.536	9.100	6.306
5 hr.	**7.570**	**4.885**	9.120	5.914	8.902	5.735	9.175	5.889	9.206	6.226	9.161	5.971	9.889	6.779
6 hr.	**8.160**	**5.250**	9.580	6.206	9.481	6.112	9.874	6.340	9.572	6.485	9.648	6.288	10.665	7.264
Average	**6.318**	**4.126**	8.369	5.459	7.427	4.819	7.743	4.999	8.504	5.805	7.942	5.201	8.415	5.891

ures 2 and 3 in the main paper, respectively. As mentioned in Section 5.2, MULTIAIR outperforms all baseline methods across all lead times, in terms of both classification and regression metrics.

Appendix B. Handling of Missing Values

In the case of time series data, missing values at each station $s \in S_T \cup S_C$ at $t' \in \{t - T_t, ..., t\}$ are filled with valid observations from the nearest station at t'. In this case, to distinguish between actual observed values and imputed values, a missing value indicator $\boldsymbol{m}_s^{(t')} \in \{0, 1\}^{d_t}$ is introduced, where d_t denotes the number of observed values in a ground observation sensor. This multi-hot vector indicates which values among $PM_{2.5}$ value and other observation values are originally missing for each station s at time t'

and is used in constructing the input time series data. For AOD images, missing values do not occur independently like ground observations but can appear simultaneously over a wide area, making the proposed imputation method inaccurate. Thus, we fill the missing values with -1, as it explicitly falls outside the typical AOD range (about 0 to 4).

Appendix C. Additional Experimental Settings

In this section, we introduce detailed settings for training MultiAir and baseline methods in Section 5.1, including hyperparameters.

For pre-training image models DeepRaNE and SimVP, we followed the training scheme of ASOC+ (Ko et al., 2022). We used Mean Squared Error (MSE) loss and the Adam (Kingma and Ba, 2015) optimizer with a learning rate 2×10^{-5}. We set the batch size to 20 and the number of training steps to 50,000. For every 1,000 steps, we evaluated the model by calculating loss on the validation set and selected the model with the best performance. The number of hidden channels of DeepRaNE was set to 32.

We used modified focal-R L1 loss in Section 4.4 with hyperparameters $\beta = 0.25$ and $\gamma = 2$ and used the Adam optimizer with a learning rate 10^{-3} for all other training processes. We set the batch size to 20 for ASOC+ and 120 for other models. We set the maximum number of epochs to 200 with early stopping if there was no performance gain for 5 epochs. For ASOC+, we set the hidden dimension for LSTM cells to 64 and the number of heads for the attention blocks to 4. For MultiAir, we set the hidden dimension for LSTM cells to 128.

Table D.3: Predictive performances on the test set in terms of extra metrics. For metrics with an upward arrow (↑), higher values correspond to better performance, whereas for those with a downward arrow (↓), lower values are preferable. In each setting, the best and the second-best performances are in **bold** and underlined.

Metric		MultiAir	ASOC+	LSTM	Transformer	TimesNet	TimeMixer	AirFormer
Classification	ACC (↑)	**0.812**	0.749	0.778	0.772	0.726	0.763	0.753
	POD (↑)	**0.728**	0.596	0.639	0.665	0.645	0.690	0.508
	FAR (↓)	0.192	0.244	0.212	0.256	0.304	0.289	**0.100**
	F1 (↑)	**0.766**	0.666	0.705	0.702	0.670	0.700	0.650
Regression	RMSE (↓)	**6.464**	8.410	7.613	7.906	8.537	8.050	8.580
	NME (↓)	**22.146**	29.346	25.907	26.871	31.208	27.959	31.670
	ρ (↑)	**0.902**	0.825	0.859	0.847	0.817	0.845	0.874

Appendix D. Evaluation Using Extra Metrics

In this section, we introduce evaluation results using extra evaluation metrics. For extra evaluation metrics, we use four classification metrics: accuracy (ACC), probability of detection (POD), false alarm rate (FAR), and F1 score, and three regression metrics: RMSE, normalized mean error (NME), and person correlation coefficient (ρ). For extra evaluation metrics, we simply evaluate once for all prediction lead times. ACC is defined as the ratio of correct class prediction based on 4 classes in Section 5.1: *Severity 1, 2, 3* and *4*. Other classification metrics, POD, FAR, and F1 score, focus on predicting extreme air pollution of *Severity 3 or worse*, which are defined as follows:

$$\text{POD} = \frac{\text{TP}}{\text{TP} + \text{FN}}, \quad \text{FAR} = \frac{\text{FP}}{\text{TP} + \text{FP}}, \quad \text{F1} = \frac{2 \times \text{TP}}{2 \times \text{TP} + \text{FP} + \text{FN}},$$

Table D.4: Ablation study results in terms of extra metrics. For metrics with an upward arrow (↑), higher values correspond to better performance, whereas for those with a downward arrow (↓), lower values are preferable. In each setting, the best and the second-best performances are in **bold** and underlined.

Metric		MultiAir	MultiAir-S	MultiAir-D	MultiAir-N	MultiAir-C
Classification	ACC (↑)	**0.812**	0.808	0.800	**0.812**	0.803
	POD (↑)	0.728	0.724	0.702	0.714	**0.739**
	FAR (↓)	0.192	0.201	0.208	**0.181**	0.226
	F1 (↑)	**0.766**	0.760	0.744	0.763	0.756
Regression	RMSE (↓)	**6.464**	6.579	6.795	6.506	6.739
	NME (↓)	22.146	22.428	23.408	**22.031**	23.036
	ρ (↑)	**0.902**	0.897	0.889	0.899	0.892

where TP, FP, FN stand for the counts of true positives, false positives, and false negatives, respectively, for air pollution of *Severity 3 or worse*. NME, ρ is defined as:

$$\text{NME} = \frac{\sum |\hat{\mathbf{P}}_T - \mathbf{P}_T|}{\sum \mathbf{P}_T}, \rho = \frac{\sum((\hat{\mathbf{P}}_T - \mu_{\hat{\mathbf{P}}_T}) \times (\mathbf{P}_T - \mu_{\mathbf{P}_T}))}{\sqrt{\sum(\hat{\mathbf{P}}_T - \mu_{\hat{\mathbf{P}}_T})^2 \times \sum(\mathbf{P}_T - \mu_{\mathbf{P}_T})^2}},$$

where $\mathbf{P}_T, \hat{\mathbf{P}}_T$ each denotes all ground truth $PM_{2.5}$ values and predicted $PM_{2.5}$ values respectively, and $\mu_{\mathbf{P}_T}$, $\mu_{\hat{\mathbf{P}}_T}$ each denotes the mean value of ground truth $PM_{2.5}$ values and predicted $PM_{2.5}$ values respectively.

Table D.3 presents the evaluation results in comparison with the baseline models. MultiAir performs the best for every metric except for FAR. However, AirFormer, which achieves the best FAR, records the lowest POD and F1 score among the models, suggesting that it simply underestimates severe air pollution.

Table D.4 presents the evaluation results in comparison with the ablation models, which are described in Section 5.3. MultiAir performs either best or second best for every metric. Especially, MultiAir consistently achieves high POD and F1 along with low FAR, indicating strong performance in extreme air pollution nowcasting.

References

Kingma, D.P., Ba, J., 2015. Adam: A method for stochastic optimization., in: International Conference on Learning Representations (ICLR).

Ko, J., Lee, K., Hwang, H., Shin, K., 2022. Deep-learning-based precipitation nowcasting with ground weather station data and radar data, in: IEEE International Conference on Data Mining Workshops (ICDMW), IEEE. pp. 1063–1070.